Assignment 1 – Cutting power and specific cutting pressure in turning, prediction of a drilling sequence

Multi-Agent System for Smart Machining (ECIU micro-module, INSA Toulouse)

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Abstract

The cutting power of the two turning programs of the reference chapter (constant cutting speed $V_c = 150\,\mathrm{m/min}$ and constant spindle speed $N = 1200\,\mathrm{rpm}$) is predicted with the mechanistic model $P_c = K_c f a_p V_c$, using the specific cutting pressure given by the chapter for the smallest diameter ($K_c = 3230.3$ MPa). The prediction is compared with the total spindle power published on GitHub, both in timing and in curve profile. A Python code then separates, pass by pass, the steady-state power during cutting from the spindle power without cutting and returns the specific cutting pressure: on the smallest diameter at constant spindle speed it gives $K_c = 3213$ MPa at $V_c = 90.5$ m/min. With these two values, the feed force, cutting power and torque of a drilling sequence (pre-drilling ∅5 mm, then drilling ∅13 mm in the same position) are predicted along time.

1 Context and data

The data come from the chapter "Integrating Environmental Impact into Cutting Speed Decisions in CNC Turning" (A. C. Araujo, document *Inspiration for your assignment*) and from its repository `github.com/annacarlaaraujo/chapter_2025`. An AISI 4135 bar is reduced from $D = 48\,\mathrm{mm}$ to 24 mm over a length of 40 mm with a DNMG coated carbide insert, in 12 passes with radial depth of cut $a_p = 1\,\mathrm{mm}$ and feed $f = 0.2\,\mathrm{mm/rev}$. Two CNC programs remove the same volume:

- **Program I**, constant cutting speed (G96), $V_c = 150\,\mathrm{m/min}$: the spindle speed grows from 1038 to 1989 rpm as the diameter decreases;
- **Program II**, constant spindle speed (G97), $N = 1200\,\mathrm{rpm}$: the cutting speed decreases from 173.4 to 90.5 m/min.

Each program was run twice, *with* chip removal and *without* ("air cutting", files `*VIDE.csv`). The Siemens trace records the spindle active power and the programmed and measured X/Z positions every 9 ms. The spindle-speed channel is empty, so the speed is taken from the program and checked through the measured feed velocity $V_f = fN$.

Figure 1 shows the programmed path. Each pass starts at $Z = +1\,\mathrm{mm}$ (1 mm of air feed before the face of the bar, $Z = 0$), feeds to $Z = -40\,\mathrm{mm}$, retracts in X, comes back in rapid and plunges to the next diameter. The programmed X is the finished diameter of the pass. As in the chapter (and in the course slides, where N is computed on the radius R_P of the tool point), the cutting speed is computed on X: $V_c = \pi X N/1000$.

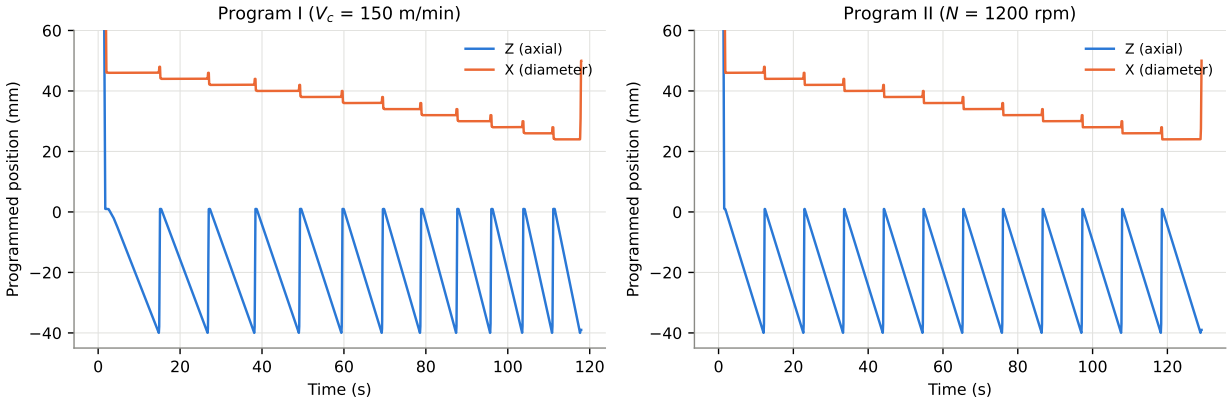


Figure 1: Programmed Z and X positions recorded during the cutting runs.

Code. All computations are in Python (environment managed with `uv`, see Appendix A): models and signal processing in `machining.py`, `turning_analysis.py` for points 1–3, `drilling_analysis.py` for point 4. `run_all.py` regenerates every figure and every number of the tables of this report.

2 Point 1 – Prediction of the cutting power in turning

Model. With the course model (slides "Power in turning")

$$F_c = K_c\, f\, a_p, \qquad P_c = F_c\, V_c = K_c\, f\, a_p\, V_c, \qquad P_f = K_f\, f\, a_p\, V_f \approx 0, \tag{1}$$

with K_c in MPa, $f\,a_p$ in mm^2 and V_c in m/s, P_c is in W. The feed power is negligible (V_f is about 1000 times smaller than V_c).

Parameters. The specific cutting pressure is the one given by the chapter (Table 2) for the *smallest diameter*, $X = 24\,\text{mm}$, in Program 2 ($N = 1200$ rpm): $K_c = 3230.3$ MPa. It is kept constant for all passes of both programs. The kinematics come from the program:

$$N_I = \frac{1000\, V_c}{\pi X}, \qquad V_{c,II} = \frac{\pi X \cdot 1200}{1000}, \qquad V_f = f\, N. \tag{2}$$

Predicted timings. For every pass (origin: start of the feed motion at $Z = +1$ mm):

- t_0, the tool touches the bar: $t_0 = 1\,\text{mm}/V_f$;
- t_1, the tool is fully engaged. With an approach angle close to 90° the whole depth of cut enters almost at once, and the transition is set by the nose radius ($r_\varepsilon = 0.8\,\text{mm}$, assumed): $t_1 = t_0 + r_\varepsilon/V_f$;
- t_2, end of the machined length: $t_2 = t_0 + 40\,\text{mm}/V_f$.

Table 1: Point 1: predicted cutting power and timings for each pass ($K_c = 3230.3$ MPa). Times measured from the start of the feed of the pass.

pass	X (mm)	N (rpm)	V_c (m/min)	V_f (mm/min)	$P_{c,pred}$ (W)	t_0 (s)	t_1 (s)	t_2 (s)
Program I – $V_c = 150$ *m/min*								
1	46	1038	150.0	207.6	1615	0.29	0.52	11.85
2	44	1085	150.0	217.0	1615	0.28	0.50	11.33
3	42	1137	150.0	227.4	1615	0.26	0.48	10.82
4	40	1194	150.0	238.7	1615	0.25	0.45	10.30
5	38	1256	150.0	251.3	1615	0.24	0.43	9.79
6	36	1326	150.0	265.3	1615	0.23	0.41	9.27
7	34	1404	150.0	280.9	1615	0.21	0.38	8.76
8	32	1492	150.0	298.4	1615	0.20	0.36	8.24
9	30	1592	150.0	318.3	1615	0.19	0.34	7.73
10	28	1705	150.0	341.0	1615	0.18	0.32	7.21
11	26	1836	150.0	367.3	1615	0.16	0.29	6.70
12	24	1989	150.0	397.9	1615	0.15	0.27	6.18
Program II – $N = 1200$ *rpm*								
1	46	1200	173.4	240.0	1867	0.25	0.45	10.25
2	44	1200	165.9	240.0	1786	0.25	0.45	10.25
3	42	1200	158.3	240.0	1705	0.25	0.45	10.25
4	40	1200	150.8	240.0	1624	0.25	0.45	10.25
5	38	1200	143.3	240.0	1543	0.25	0.45	10.25
6	36	1200	135.7	240.0	1461	0.25	0.45	10.25
7	34	1200	128.2	240.0	1380	0.25	0.45	10.25
8	32	1200	120.6	240.0	1299	0.25	0.45	10.25
9	30	1200	113.1	240.0	1218	0.25	0.45	10.25
10	28	1200	105.6	240.0	1137	0.25	0.45	10.25
11	26	1200	98.0	240.0	1055	0.25	0.45	10.25
12	24	1200	90.5	240.0	974	0.25	0.45	10.25

Program I gives a constant cutting power, $P_c = 1615$ W, on passes that become shorter (11.6 to 6.0 s) because V_f grows with N. Program II gives passes of constant duration (10 s of cut) with a cutting power proportional to the diameter, decreasing linearly from 1867 to 974 W.

3 Point 2 – Recorded total power: timings and curve profile

Figure 2 shows the total spindle power recorded on GitHub, with and without chip removal. The machine does not measure P_c alone: the predicted *total* power is therefore built as $P_{tot,pred}(t) = P_0 + P_c(t)$, where P_0 is the spindle power without cutting identified on the air-cutting run (Section 4) and $P_c(t)$ follows Eq. (1) with a ramp from t_0 to t_1. It is overlaid on the measured signal in Fig. 3.

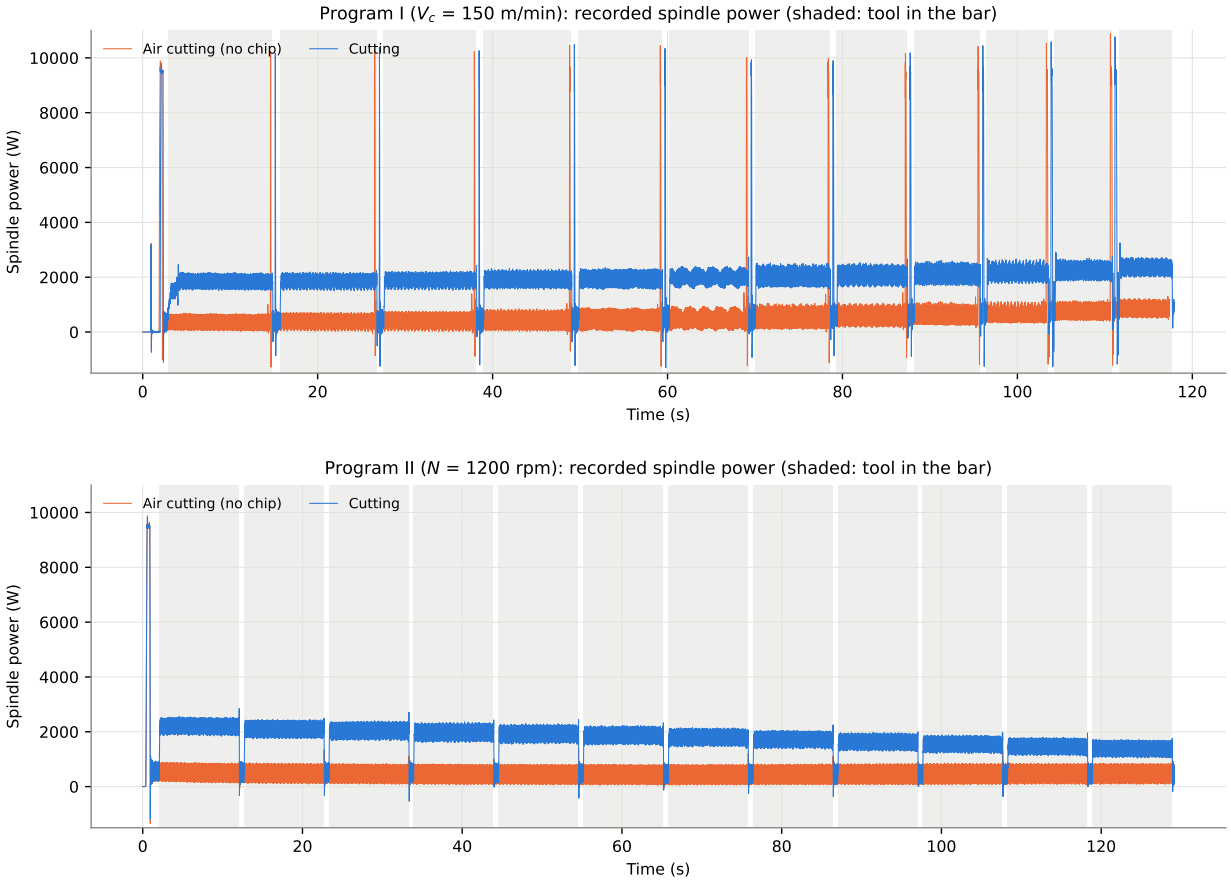


Figure 2: Recorded spindle power with chip removal (blue) and without (orange). Shaded bands: tool inside the bar ($0 \geq Z \geq -40$ mm).

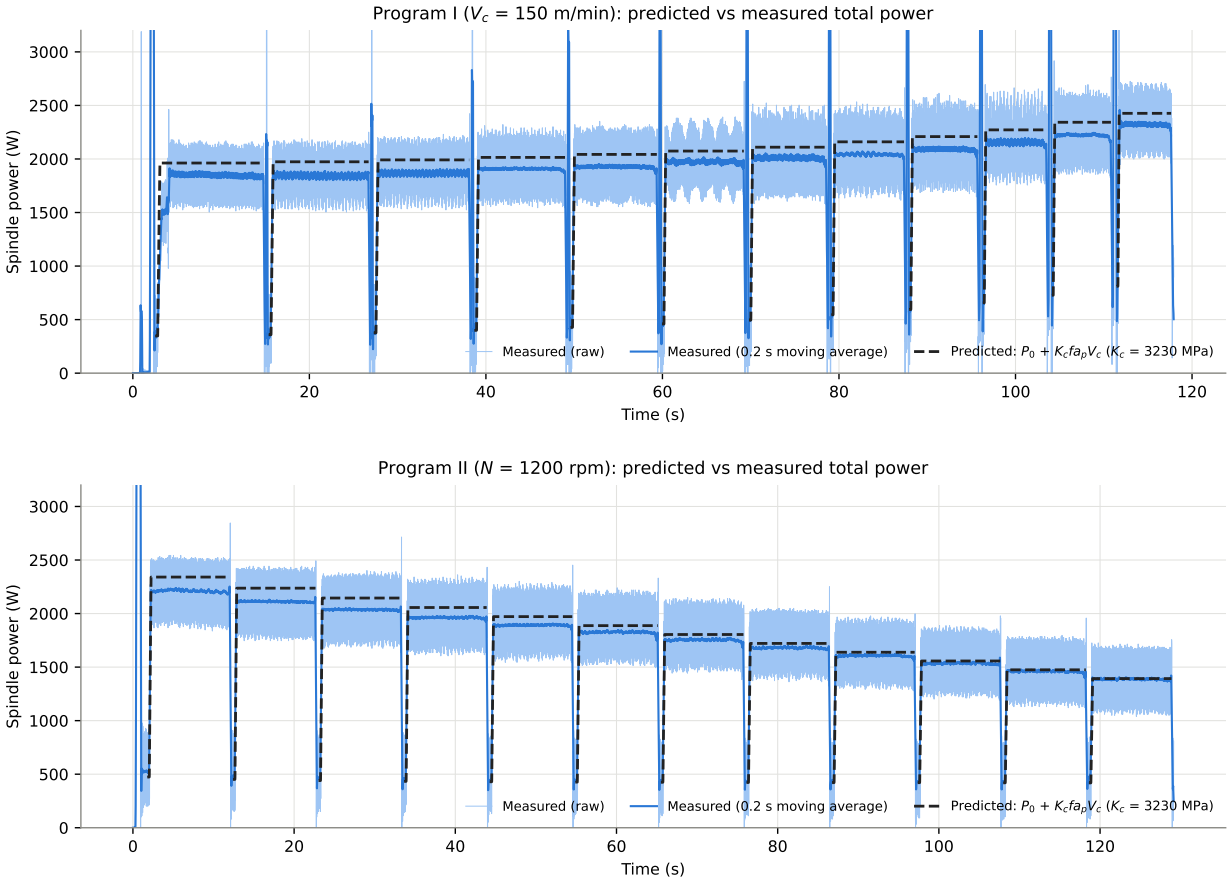


Figure 3: Predicted total power (P_0 + mechanistic P_c with $K_c = 3230.3$ MPa) against the measured power (raw and 0.2 s moving average).

3.1 Timings

Table 2 compares, pass by pass, the predicted feed velocity and cutting time with the measured ones. The measured feed velocity is the slope of the recorded $Z(t)$. The measured cutting time is the time during which the smoothed power stays above $P_0 + 0.5\,P_c$.

- The measured V_f equals the predicted $f\,N$ to 0.01 mm/min: in G96 the controller applies exactly $N = 1000V_c/(\pi X)$.
- The cutting time read on the power signal matches $40/V_f$ within 0.2 %. In total, 105.6 s are measured against 105.6 s predicted for Program I, and 119.8 s against 120.0 s for Program II. Program I is faster (total cycle 117.9 s against 129.1 s) because the feed velocity increases on the small diameters.
- The power rises 0.07–0.09 s after the predicted contact t_0 and falls about 0.06 s after t_2. The delay is the same at entry and at exit, so it is a lag of the power measurement (drive filtering), not a geometric effect. The 10–90 % rise time is about 0.1 s (about 2 revolutions), a little faster than the nose-radius estimate $t_1 - t_0 = r_\varepsilon/V_f \approx 0.2$ s (Fig. 4).
- Pass 1 of Program I is the only exception. The spindle is still accelerating when the tool enters: the feed per revolution is respected but V_f is lower for about 1.4 s, so the contact is late and the power ramp lasts 1 s.

Table 2: Timing comparison per pass. Δt_0: time from the start of the pass to the contact (predicted) and to the power rise (measured). E: energy of the cutting run during the cut; SEC $= E/V_{removed}$.

	X	V_f (mm/min)		Δt_0 (s)		T_{cut} (s)		rise	E	SEC
pass	(mm)	pred.	meas.	pred.	meas.	pred.	meas.	(s)	(kJ)	(J/mm^3)
Program I – $V_c = 150$ m/min										
1	46	207.6	207.6	0.29	0.56	11.56	11.77	1.05	21.8	3.69
2	44	217.0	217.0	0.28	0.35	11.06	11.03	0.09	20.3	3.60
3	42	227.4	227.4	0.26	0.33	10.56	10.54	0.12	19.7	3.64
4	40	238.7	238.7	0.25	0.31	10.05	10.03	0.10	19.2	3.72
5	38	251.3	251.3	0.24	0.31	9.55	9.53	0.10	18.4	3.75
6	36	265.3	265.3	0.23	0.30	9.05	9.04	0.10	17.8	3.84
7	34	280.9	280.9	0.21	0.29	8.55	8.52	0.09	17.1	3.90
8	32	298.4	298.4	0.20	0.27	8.04	8.03	0.08	16.4	3.96
9	30	318.3	318.3	0.19	0.26	7.54	7.52	0.08	15.7	4.04
10	28	341.0	341.0	0.18	0.24	7.04	7.02	0.07	15.1	4.15
11	26	367.3	367.3	0.16	0.23	6.53	6.52	0.23	14.5	4.27
12	24	397.9	397.9	0.15	0.22	6.03	6.02	0.21	14.0	4.45
Program II – $N = 1200$ rpm										
1	46	240.0	240.0	0.25	0.32	10.00	9.98	0.09	22.1	3.74
2	44	240.0	240.0	0.25	0.32	10.00	9.98	0.10	21.1	3.73
3	42	240.0	240.0	0.25	0.32	10.00	9.98	0.09	20.3	3.76
4	40	240.0	240.0	0.25	0.31	10.00	9.99	0.10	19.6	3.81
5	38	240.0	240.0	0.25	0.31	10.00	9.98	0.09	18.9	3.85
6	36	240.0	240.0	0.25	0.32	10.00	9.99	0.09	18.3	3.93
7	34	240.0	240.0	0.25	0.31	10.00	9.98	0.08	17.5	3.98
8	32	240.0	240.0	0.25	0.32	10.00	9.97	0.10	16.8	4.04
9	30	240.0	240.0	0.25	0.31	10.00	9.99	0.09	16.1	4.12
10	28	240.0	240.0	0.25	0.32	10.00	9.99	0.08	15.3	4.21
11	26	240.0	240.0	0.25	0.31	10.00	9.99	0.10	14.6	4.30
12	24	240.0	240.0	0.25	0.31	10.00	10.00	0.10	13.9	4.42

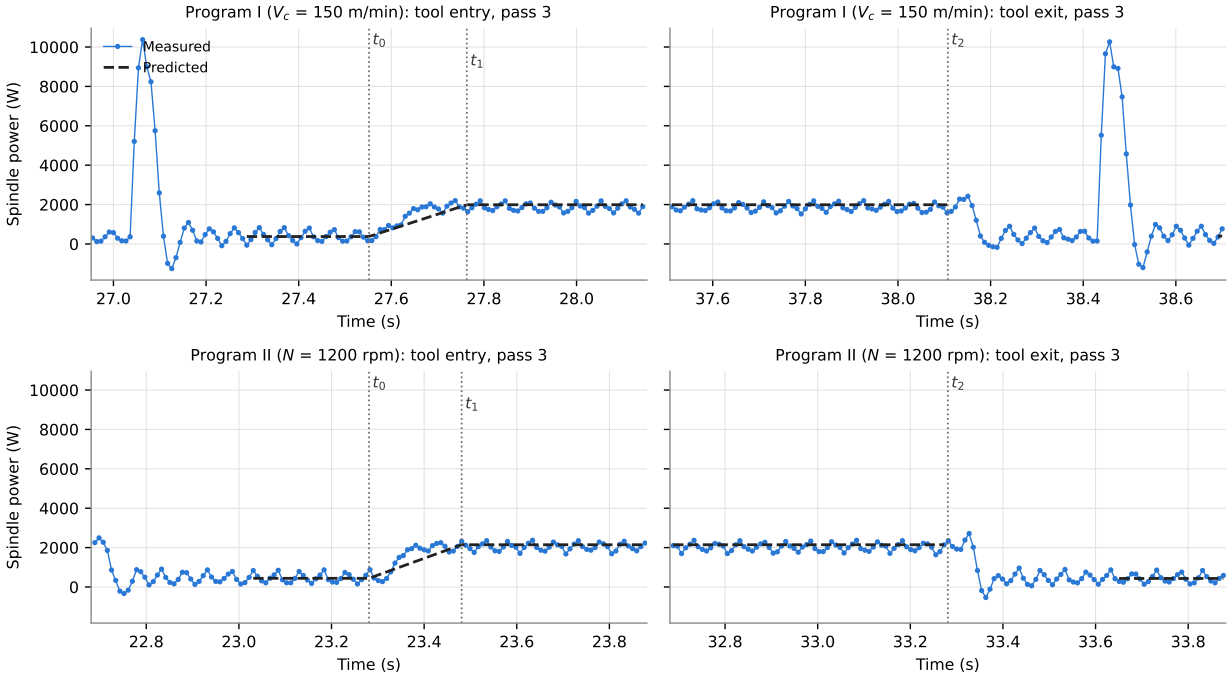


Figure 4: Tool entry and exit (pass 3): measured samples and prediction.

3.2 Curve profile: conclusions

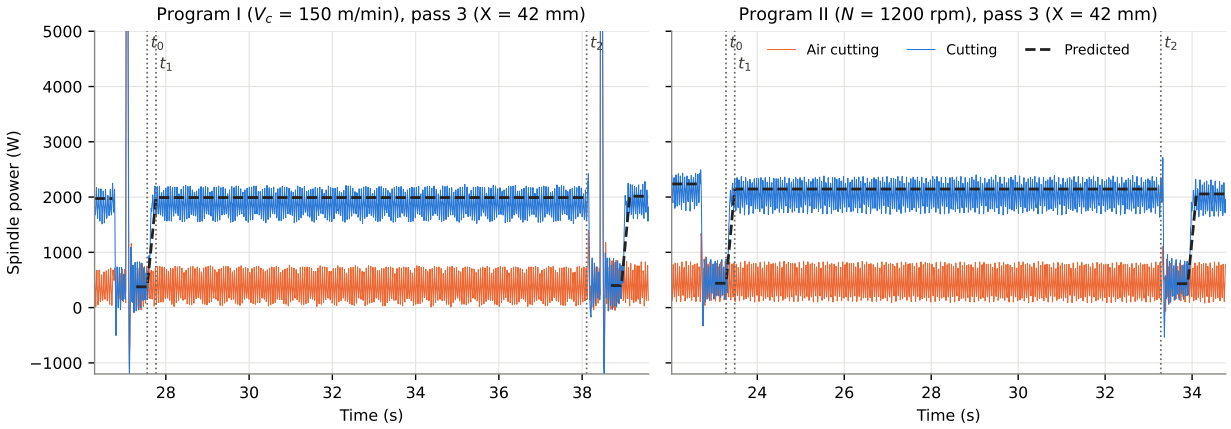


Figure 5: Pass 3 of both programs: cutting run, air-cutting run (re-aligned on the same pass) and prediction.

1. **Plateaus.** During each pass the power is a flat plateau, as predicted: f, a_p and V_c do not change along Z. The plateau level is $P_0 + P_c$. The recorded curve is the sum of the mechanistic cutting power and of the power needed to keep the spindle rotating.

2. **Program I.** P_c is constant (1502 W measured), but the total power *increases* pass after pass (1851 to 2323 W) because P_0 grows with the spindle speed (347 to 811 W). The rise comes from the spindle, not from the cut. The plateaus get shorter because $V_f = fN$ increases.

3. **Program II.** P_0 is almost constant and $P_c \propto V_c \propto X$ decreases linearly (1738 to 969 W), so the total power *decreases* pass after pass, with plateaus of constant length.

4. **Level of the prediction.** With the constant $K_c = 3230.3$ MPa, the predicted cutting power is 6.6–8.8 % too high in Program I. In Program II the error is 7.6 % on the first passes (high V_c) and falls to 0.6 % on the last one, the pass on which K_c was obtained. The error grows with the cutting speed: K_c is not exactly constant but decreases when V_c increases (Section 4). The *shape* and the *timings* of the curves are reproduced exactly.

5. **Peaks between passes.** In Program I each plunge to a smaller diameter makes the G96 controller accelerate the spindle, which gives peaks of about 10 kW (inertial power $J\omega\dot{\omega}$). They are preceded by negative values when the tool retracts in X and the spindle is braked (regenerative drive). In Program II the speed never changes, so the only large peak is the initial spindle start. These transients are outside the cutting model but consume energy.

6. **Ripple.** The raw signal oscillates by ±200 W to 300 W around the plateau, with and without cutting, at the rotation frequency $N/60$ (20.0 Hz at 1200 rpm, Fig. 6). It is a once-per-revolution effect of the rotating assembly (unbalance or eccentricity of chuck and bar), not a cutting phenomenon, and it averages out over the steady-state window.

7. **Energy.** Despite the different profiles, the energy spent during the cut is almost the same in the two programs: 210 kJ against 214 kJ in total, of which 159 kJ against 163 kJ for cutting. The cutting energy per unit volume equals K_c (SEC_{cut} = 2.92 and 3.01 J/mm^3, i.e. about 3000 MPa) and does not depend on the strategy; the difference lies in the non-productive spindle energy.

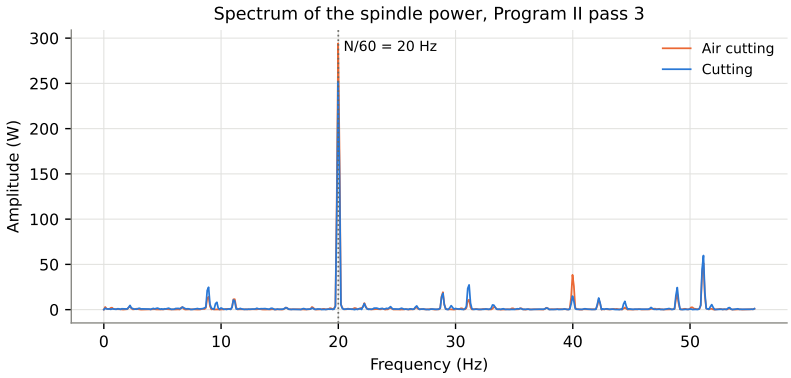


Figure 6: Amplitude spectrum of the spindle power in steady state (Program II, pass 3): the ripple is at the rotation frequency $N/60$.

4 Point 3 – Specific cutting pressure from the recorded power

4.1 Method

For every pass the code:

1. finds the feed motions from the derivative of the programmed Z (feed between 30 and 1000 mm/min). This pairs the cutting run and the air-cutting run pass by pass, even though the two runs are shifted in time;

2. takes the *steady-state window* $-3 \geq Z \geq -37$ mm: the tool is fully engaged and far from the entry/exit transients and from the acceleration peaks. The window covers tens of revolutions, so the once-per-revolution ripple averages out;

3. computes P_{tot}, the mean power of the cutting run, and P_0, the mean power of the air-cutting run in the same window (spindle power without cutting, same speed, same axis motions);

4. computes the cutting power and the specific cutting pressure (chapter Eqs. 5–6):

$$P_c = P_{tot} - P_0, \qquad F_c = \frac{P_c}{V_c}, \qquad K_c = \frac{P_c}{f\, a_p\, V_c}. \tag{3}$$

```
p_tot, _ = steady_state_mean(cut, pc.i_start, pc.i_end, Z_HI, Z_LO) # cutting run
p_air, _ = steady_state_mean(air, pa.i_start, pa.i_end, Z_HI, Z_LO) # air cutting
p_cut = p_tot - p_air
Kc = kc_turning(p_cut, F, AP, Vc) # = p_cut / (F*AP*Vc/60)
```

4.2 Results

Table 3: Program I (V_c = 150 m/min): steady-state powers (W), cutting force (N) and specific cutting pressure (MPa), compared with Table 2 of the chapter.

pass	X	N	V_c	P_{tot}	P_0	P_c	F_c	K_c	K_c chapter
1	46	1038	150.0	1851	347	1504	601	3007	3012
2	44	1085	150.0	1843	359	1485	594	2969	2968
3	42	1137	150.0	1868	376	1492	597	2984	2985
4	40	1194	150.0	1908	400	1509	604	3018	3021
5	38	1256	150.0	1927	428	1499	600	2998	2998
6	36	1326	150.0	1973	459	1515	606	3029	3025
7	34	1404	150.0	2011	495	1516	606	3031	3035
8	32	1492	150.0	2043	545	1499	599	2997	2998
9	30	1592	150.0	2089	594	1496	598	2991	2994
10	28	1705	150.0	2157	657	1500	600	3000	3003
11	26	1836	150.0	2224	728	1496	598	2992	2996
12	24	1989	150.0	2323	811	1512	605	3023	3026

Table 4: Program II (N = 1200 rpm): same quantities as Table 3. The last row is the reference used for drilling.

pass	X	N	V_c	P_{tot}	P_0	P_c	F_c	K_c	K_c chapter
1	46	1200	173.4	2212	474	1738	601	3007	3020
2	44	1200	165.9	2111	451	1660	601	3003	3000
3	42	1200	158.3	2036	440	1596	605	3025	3028
4	40	1200	150.8	1963	433	1530	609	3044	3047
5	38	1200	143.3	1892	429	1463	613	3064	3069
6	36	1200	135.7	1827	426	1401	619	3097	3088
7	34	1200	128.2	1755	424	1330	623	3114	3118
8	32	1200	120.6	1681	422	1259	626	3131	3121
9	30	1200	113.1	1608	421	1187	630	3149	3150
10	28	1200	105.6	1535	421	1114	633	3166	3179
11	26	1200	98.0	1461	419	1041	637	3187	3190
12	24	1200	90.5	1387	419	969	643	3213	3230

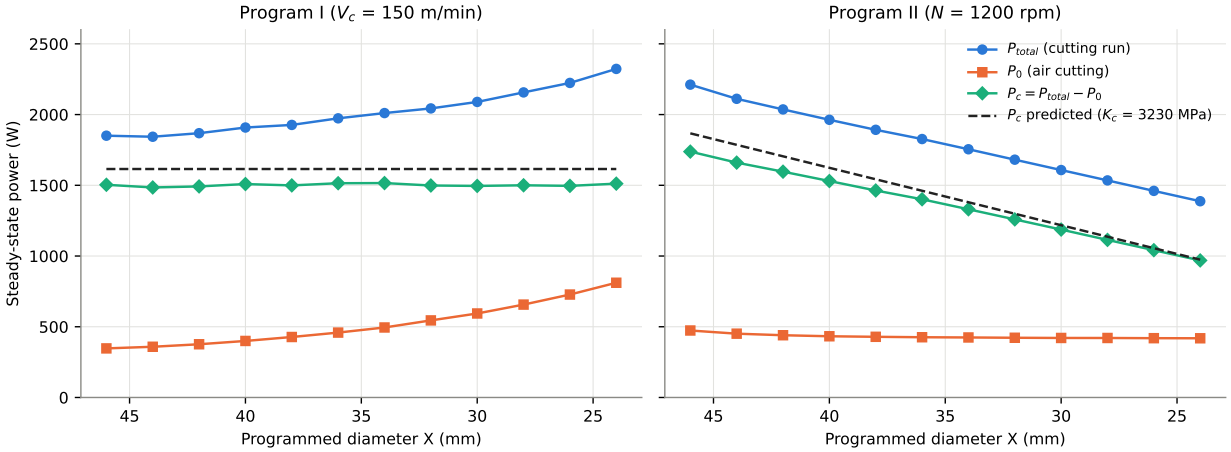


Figure 7: Steady-state powers per pass: total, without cutting, cutting power (identified and predicted at point 1).

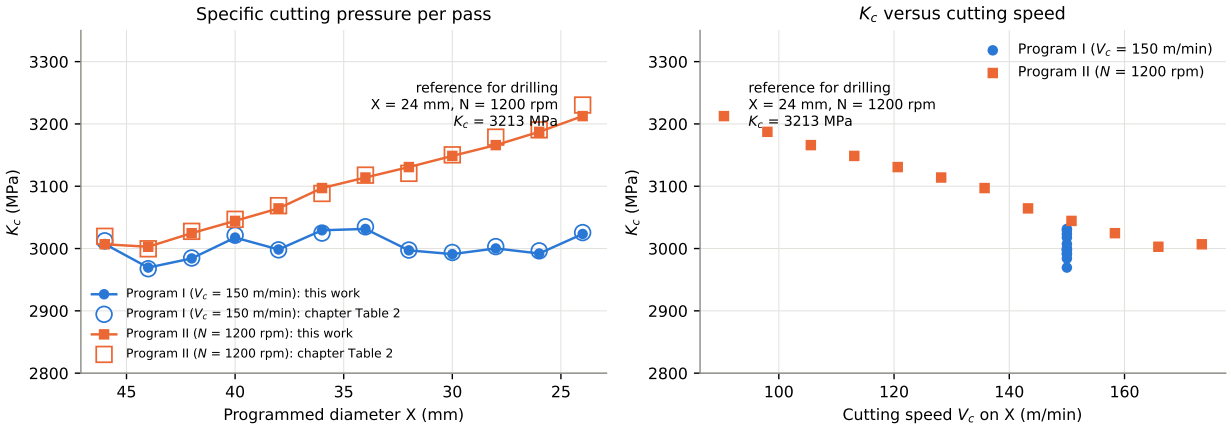


Figure 8: Specific cutting pressure identified on every pass, compared with the chapter.

Discussion.

- The values reproduce Table 2 of the chapter within 0.5 %. Program I: $K_c = 3003 \pm 19$ MPa. Program II: K_c from 3003 to 3213 MPa.
- **Reference value for point 4** (smallest diameter, constant spindle speed, $X = 24\,\text{mm}$, $N = 1200\,\text{rpm}$): $P_c = 969$ W, $F_c = 643$ N, $V_c = 90.5$ m/min, $K_c = 3213$ MPa. The chapter gives 3230.3 MPa because it uses the rounded speed $V_c = 90\,\text{m/min}$; with the exact speed $\pi \cdot 24 \cdot 1200/1000 = 90.48\,\text{m/min}$ the same power gives 3213 MPa.
- In Program II, K_c increases when V_c decreases (from 173 to 90 m/min, Fig. 8 right). This is the known decrease of the cutting force with the cutting speed at moderate speeds (softening of the shear zone, see the "Experiments" slide). In Program I, where V_c is constant, K_c is constant too. A constant-K_c model is therefore exact only at the speed where K_c was measured, which explains the deviations found at point 2.
- P_0 follows a quadratic law of the speed in Program I, $P_0 \approx 55 + 0.156\,N + 1.14 \cdot 10^{-4} N^2$ (W, rpm), Fig. 9. In Program II, at constant 1200 rpm, P_0 decreases from 474 to 419 W and tends to the value 406 W of the Program I fit: the spindle is warming up (viscosity of the bearing lubricant). Since P_0 is measured in a separate run, a different thermal state between the cutting and air-cutting runs is the main uncertainty of the method: $\pm 30\,\text{W}$ on P_0 gives about $\pm 100\,\text{MPa}$ on K_c at 90 m/min.

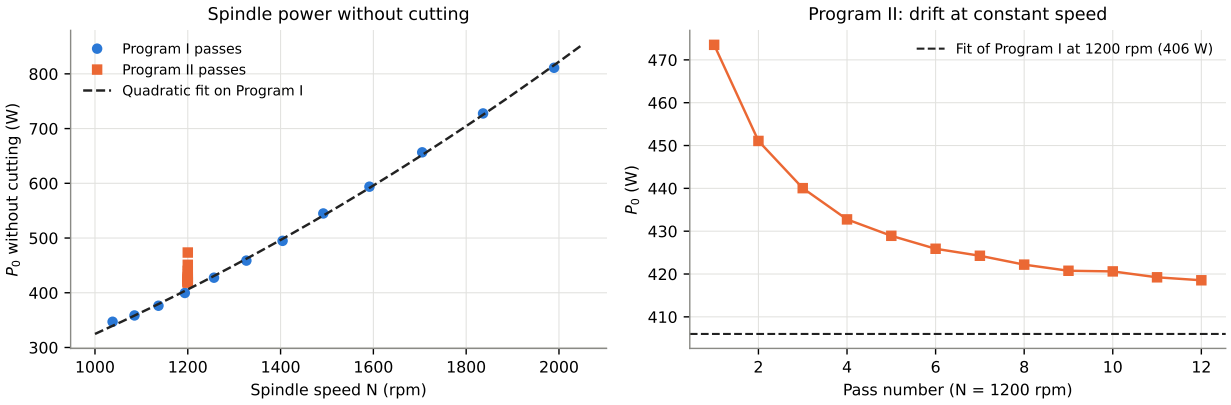


Figure 9: Spindle power without cutting: dependence on the spindle speed and thermal drift at constant speed.

4.3 Minimum and maximum values

The recorded power is noisy: around each plateau it oscillates once per revolution (Fig. 6). Besides the mean, the minimum and maximum of the power were therefore taken in the same steady-state window, for the cutting run and for the air-cutting run. To avoid that a single outlier sample sets the value, the minimum and maximum are the 1 % and 99 % percentiles of the samples of the window. The cutting power and the specific cutting pressure are then computed from the minima and from the maxima:

$$P_c^{min} = P_{tot}^{min} - P_0^{min}, \qquad P_c^{max} = P_{tot}^{max} - P_0^{max}, \qquad K_c^{min,max} = \frac{P_c^{min,max}}{f\, a_p\, V_c}. \tag{4}$$

The minimum of the cutting run is paired with the minimum of the air-cutting run, and the maximum with the maximum, because the oscillation comes from the rotation and is present in both runs.

Table 5: Minimum and maximum steady-state powers (W) and the resulting specific cutting pressure (MPa), compared with the value from the mean powers.

	X	P_{tot}		P_0		P_c		K_c		
pass	(mm)	min	max	min	max	from min	from max	from min	mean	from max
Program I – $V_c = 150$ *m/min*										
1	46	1534	2160	50	666	1484	1494	2967	3007	2987
2	44	1524	2160	40	707	1484	1453	2967	2969	2907
3	42	1544	2200	30	747	1514	1453	3028	2984	2907
4	40	1574	2286	40	823	1534	1463	3068	3018	2927
5	38	1564	2301	40	878	1524	1423	3048	2998	2846
6	36	1605	2382	61	929	1544	1453	3088	3029	2907
7	34	1635	2482	101	1019	1534	1463	3068	3031	2926
8	32	1645	2483	182	1029	1463	1453	2927	2997	2907
9	30	1696	2564	252	1069	1443	1495	2887	2991	2990
10	28	1776	2624	333	1110	1443	1514	2887	3000	3028
11	26	1859	2644	424	1120	1435	1524	2870	2992	3048
12	24	2008	2708	515	1211	1494	1497	2987	3023	2994
Program II – $N = 1200$ *rpm*										
1	46	1867	2523	151	874	1716	1649	2968	3007	2854
2	44	1756	2432	121	848	1635	1585	2957	3003	2866
3	42	1696	2372	95	828	1600	1544	3032	3025	2926
4	40	1625	2331	81	818	1544	1514	3072	3044	3012
5	38	1564	2271	71	807	1494	1463	3128	3064	3065
6	36	1524	2220	71	797	1453	1423	3213	3097	3146
7	34	1463	2130	61	797	1403	1332	3284	3114	3118
8	32	1383	2039	71	818	1312	1221	3263	3131	3037
9	30	1302	1958	71	838	1231	1120	3266	3149	2972
10	28	1221	1867	81	848	1140	1019	3241	3166	2897
11	26	1140	1782	91	838	1050	944	3213	3187	2890
12	24	1060	1696	91	838	969	858	3213	3213	2845

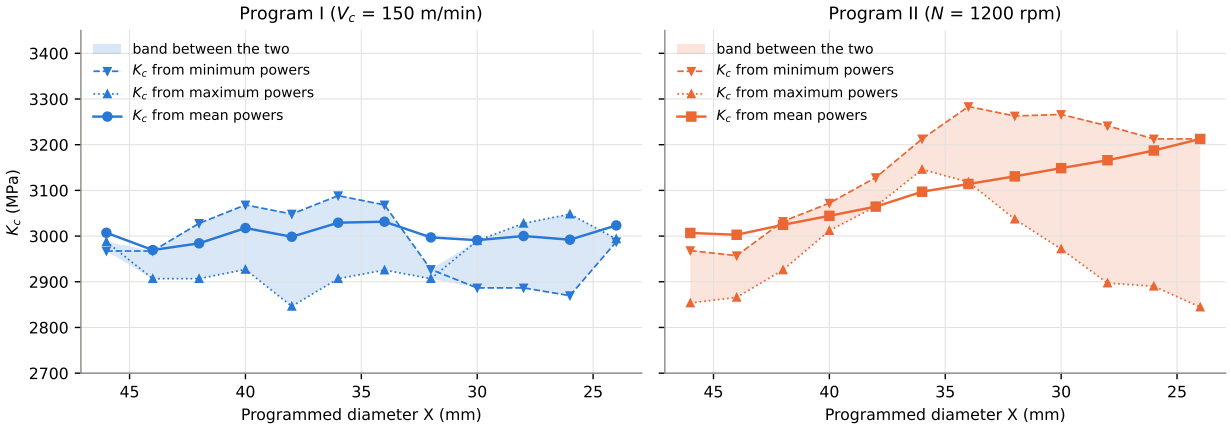


Figure 10: Specific cutting pressure from the mean, minimum and maximum powers.

- Within a pass the power swings by about 710 W with cutting and 756 W without cutting, i.e. by the same amount. The oscillation is not produced by the cut but by the rotating assembly, and it is simply added to the cutting power.
- For this reason P_c from the minima and P_c from the maxima stay close to P_c from the means. All the K_c values fall between 2845 and 3284 MPa, at most 11.5 % from the value obtained from the means. The band is at most 202 MPa wide in Program I and 368 MPa in Program II. In both cases it is wider than the change of K_c from one pass to the next.
- On the reference pass (X = 24 mm, N = 1200 rpm), K_c is 3213 MPa from the minima, 3213 MPa from the means and 2845 MPa from the maxima.
- Pairing the values the other way round ($P_{tot}^{min} - P_0^{max}$ and $P_{tot}^{max} - P_0^{min}$) would give, on the same pass, K_c between 736 and 5321 MPa. This range has no physical meaning: it only adds up the two oscillations, which happen together.
- The minimum and maximum come from a few samples at the extremes of the oscillation, while the mean uses all the samples of the window. The mean is therefore the most robust estimate, and it is the value used for the drilling prediction.

5 Point 4 – Prediction of the drilling sequence

5.1 Data and model

The specific cutting pressure and the cutting speed of point 3 are used: K_c = 3213 MPa, $K_f = 0.4\,K_c$ = 1285 MPa and V_c = 90.5 m/min (peripheral speed of the drill). The sequence is:

- pre-drilling ∅5 mm, point angle 120°, machining length 10 mm, f = 0.2 mm/rev: N = 5760 rpm, V_f = 1152 mm/min;
- drilling ∅13 mm in the same position, point angle 120°, machining length 20 mm, f = 0.2 mm/rev: N = 2215 rpm, V_f = 443 mm/min.

The machining length is taken as the depth of the drill tip below the part surface at the end of the feed. The tool starts 1 mm above the surface; times are measured from the start of the feed.

Following the course slides (“Machining forces in drilling”), each of the two lips cuts with a feed per tooth $f_z = f/2$ and a cutting speed that grows linearly with the radius, $V_c(r) = 2\pi r N/1000$ (Fig. 11). Each lip is divided into elements dr:

$$dF_c = K_c\, f_z\, dr, \qquad dF_f = K_f\, f_z\, dr, \qquad dP_c = dF_c\, V_c(r), \tag{5}$$

summed over the elements that are in the material, for both lips. With the drill tip at depth z, an element at radius r sits at $z - r\cot 60°$ and cuts if it is below the surface and not inside the pilot hole. On the full lip this gives the formulas of the slides:

$$P_c = 2\,K_c\,\frac{D}{2}\,\frac{f}{2}\,\frac{V_c}{2} = K_c\,\frac{\pi D^2}{4}\,fN, \qquad F_f = K_f\,\frac{D}{2}\,f, \qquad M_c = \frac{P_c}{\omega} = K_c\,\frac{D^2 f}{8}. \tag{6}$$

When enlarging a pilot hole of diameter d the lips only cut the annulus $d/2 \le r \le D/2$:

$$P_c = K_c\,\frac{\pi(D^2 - d^2)}{4}\,fN, \qquad F_f = K_f\,\frac{D-d}{2}\,f. \tag{7}$$

The chisel edge is neglected (it adds thrust force but little power), and so is the spindle power without cutting: $P_0(N)$ was only measured up to 2000 rpm, and extrapolating it to the 5760 rpm of the ⌀5 drill would not be reliable.

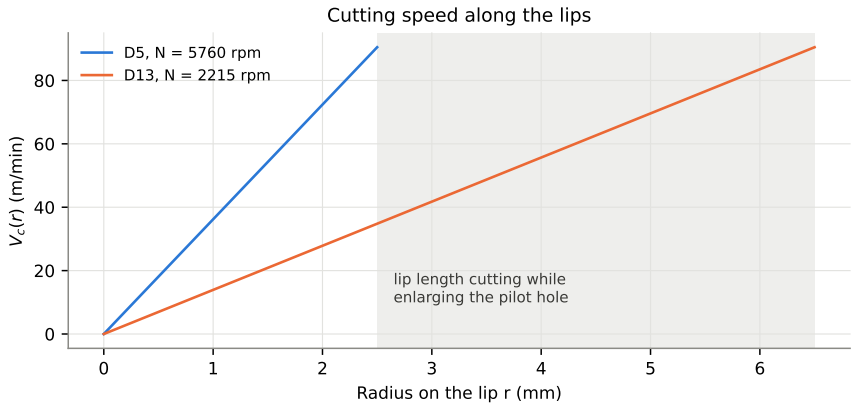


Figure 11: Cutting speed along the lips. Shaded: part of the ⌀13 lip that cuts while the pilot hole is being enlarged.

5.2 Characteristic instants

Pre-drilling ⌀5:

- $t_0 = 0.052$ s: the tip touches the part ($z = 0$);
- $t_1 = 0.127$ s: the conical point (height $2.5/\tan 60° = 1.44$ mm) is fully inside the material;
- $t_2 = 0.573$ s: end of the machining length ($z = 10\,\text{mm}$).

Drilling ⌀13 inside the pilot hole:

- $t_0 = 0.331$ s: the lips touch the edge of the pilot hole, at $r = 2.5\,\text{mm}$, when the tip is at $z = 1.44$ mm. Before that, the tip moves in the empty pilot hole;
- $t_1 = 0.644$ s: the whole annulus $2.5 \le r \le 6.5$ mm is engaged ($z = 3.75$ mm), and the hole is being enlarged;
- $t_2 = 1.490$ s: the tip reaches the bottom of the pilot hole ($z = 10\,\text{mm}$). The bottom of the pilot hole is a cone with the same 120° angle as the drill, so the inner part of the lips ($r < 2.5$ mm) engages *all at once*: power, force and torque step up. In practice the chisel edge also starts working here, which makes the thrust step even larger;
- $t_3 = 2.844$ s: end of the machining length ($z = 20\,\text{mm}$), full diameter drilling between t_2 and t_3.

5.3 Predicted curves

Table 6: Steady values of each phase of the drilling sequence (K_c = 3213 MPa, K_f = 1285 MPa, V_c = 90.5 m/min).

drill	phase	cutting radii (mm)	F_f (N)	P_c (W)	M_c (N m)	P_f (W)
D5	t_1–t_2	0.0–2.5	643	1211	2.01	12.3
D13	t_1–t_2	2.5–6.5	1028	2683	11.57	7.6
D13	t_2–t_3	0.0–6.5	1671	3149	13.57	12.3

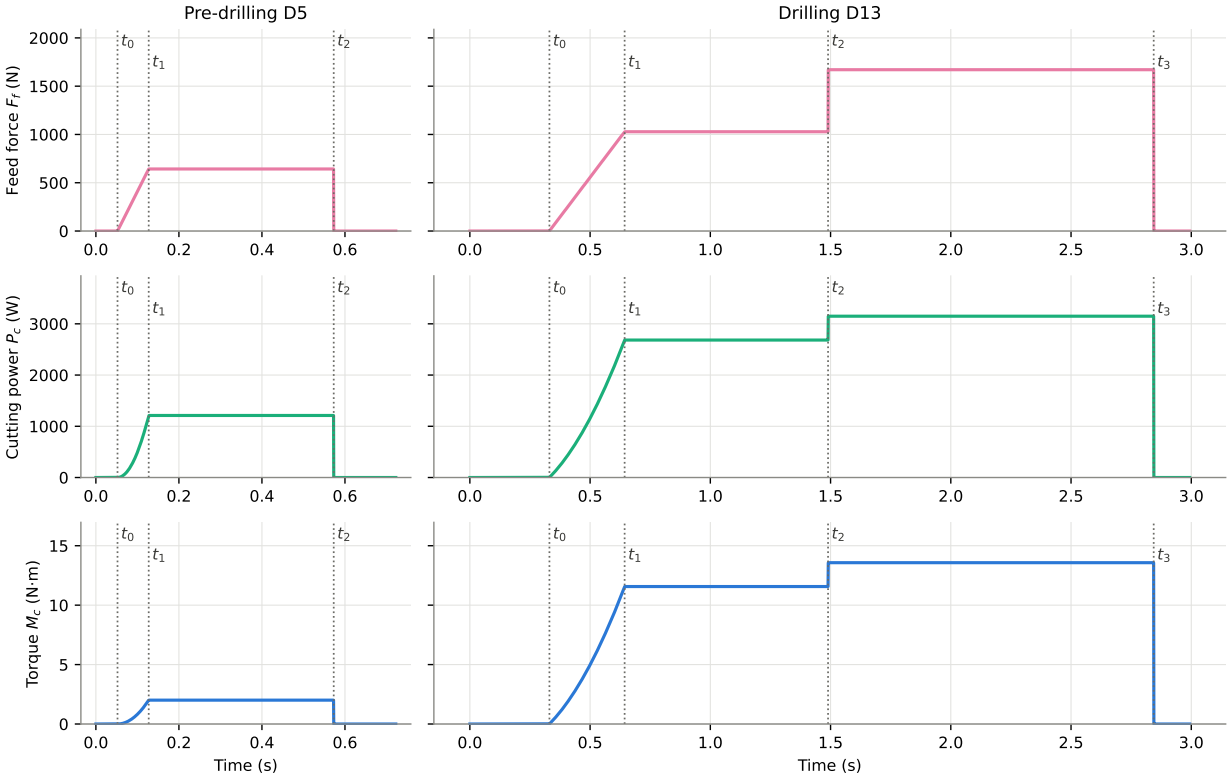


Figure 12: Predicted feed force, cutting power and torque of the two operations, on the same vertical scales.

Curve profile.

- **Entry ($t_0 \to t_1$).** The engaged lip length grows linearly with the penetration, so the feed force grows *linearly* in time. Power and torque also depend on the radius of the engaged elements ($V_c \propto r$), so they grow *quadratically* for the ∅5 drill. For the ∅13 drill they start from $r = 2.5$ mm, where V_c is already 35 m/min, so the curve is closer to a straight line.
- **Pilot hole.** The pilot hole cuts the thrust force of the ∅13 drill by 38 % during the enlargement (1028 against 1671 N). The cutting power drops by only 15 % (2683 against 3149 W) because the removed core is the part of the lip with the lowest cutting speed. This is why pilot holes are used to reduce the axial load.
- **Comparison of the two drills.** With the same V_c and the same f, the ∅13 drill turns 2.6 times slower and V_f is 2.6 times lower. Its power is 2.6 times higher (proportional to D) and its torque 6.8 times higher (proportional to D^2).
- **Energy.** The cutting energy is $E = \int P_c\,dt = K_c\,V_{removed}$: 570 J for the pre-drilling and 6892 J for the drilling.

- **Limits.** K_c was measured in turning with an uncut chip thickness $h = f \sin \kappa_r \approx 0.2\,\text{mm}$. In drilling $h = f_z \sin 60° \approx 0.087\,\text{mm}$, and the speed varies from 0 to 90 m/min along the lip. Both effects (size effect and low speed near the centre, Fig. 8) would increase the real K_c, so the predicted curves are a lower bound. The chisel edge, the friction of the margins and the chip evacuation in the 20 mm-deep hole (see the torque curves of Davim in the slides) are not modelled either. The total power read by the machine would also include $P_0(N)$.

6 Conclusions

- With $K_c = 3230.3$ MPa (chapter, smallest diameter) the mechanistic model predicts the timings of every pass within 0.2 %. It overestimates the cutting power by up to about 9 % at high cutting speed, because K_c decreases when V_c increases.
- The recorded total power is the sum of a spindle term $P_0(N)$ and of the cutting term: in Program I the total power rises because of the spindle, in Program II it falls because of the cut. Acceleration peaks of about 10 kW appear only when the spindle speed changes (Program I). The power signal lags the position by about 0.07 s.
- The Python code separates the steady-state cutting power from the spindle power without cutting and finds K_c for every pass within 0.5 % of the chapter. On the smallest diameter at constant spindle speed, $K_c = 3213$ MPa at $V_c = 90.5$ m/min. Computed from the minimum and maximum powers instead of the means, K_c stays between 2845 and 3284 MPa, because the power oscillation is the same with and without cutting.
- Drilling sequence: the ∅5 pre-drilling needs about 1.2 kW, 643 N and 2.0 N m. The ∅13 drilling needs 2.7 kW and 1.0 kN while it enlarges the pilot hole, then steps up to 3.1 kW, 1.7 kN and 13.6 N m when the tip reaches the bottom of the pilot hole at t_2.

A Code

```
Assigment1/
  pyproject.toml, uv.lock # uv environment (numpy, pandas, matplotlib, scipy)
  run_all.py # runs everything (turning first, then drilling)
  data/ # GitHub traces (chapter_2025 repository)
  src/machining.py # models, trace reader, pass detection, steady state
  src/turning_analysis.py # points 1-3 (turning), exports the reference Kc
  src/drilling_analysis.py # point 4 (drilling sequence D5 + D13)
  src/plotstyle.py # common figure style
  plot/ # all figures (PDF + PNG)
  report/ # this report (+ generated/ tables and numbers)
```

To reproduce:

```
uv sync
uv run python run_all.py
cd report && latexmk -pdf main.tex
```